What Drives the Scenario? Interpreting Conditional Forecasts in Reduced-Form VARs

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Conditional Forecasting in Modern Macroeconomic Policy

Conditional forecasts are standard in policy

Targets (e.g., inflation) conditional on assumed paths (e.g., oil, interest rate)

Binder & Sekkel (2024)

Structural models

- ✓ Coherent economic narratives (identified shocks)
- × Sensitive to identification assumptions & sample variation (Antolín-Díaz et al., 2021)

Reduced-form (B)VARs

- ✓ Are often used to cross-check macroeconomic outlooks (Angelini et al., 2025)
- ✓ Robust, easy to implement
- × No causal interpretation

Out-of-Sample Evaluation and Interpretability Limits

In reduced-form VARs, conditional forecasts are typically evaluated out of sample in two main ways (Angelini et al., 2019):

- **Realized-path evaluation.** Conditional forecasts are generated using the *realized* future values of the conditioning variables and evaluated against observed outcomes. This assesses the information content of the conditioning set.
- **Scenario-path evaluation.** Forecasts are conditioned on assumed (real-time) scenario paths and evaluated relative to the same assumed paths, providing a real-time measure of accuracy.

Limitation. Forecast accuracy alone is vulnerable to the evaluation period: inferring that a variable is "important" conflates

1. the *weight* the model assigns to that variable, and
2. the *realized dynamics* of the variable during the evaluation window.

Research Contributions

1. **A new interpretability framework for conditional forecasts.** Using the Kalman filter weights, we develop model-consistent tools that decompose conditional forecasts into contributions from specific elements of the imposed scenario paths, enabling a transparent economic interpretation of reduced-form VAR scenarios.
2. **Linking scenario analysis to impulse response theory.** We formally establish when the weights coincide with generalized impulse response functions (single-period conditioning) and when this equivalence breaks down under multi-period path restrictions.
3. **Weight-based measures of variable importance.** We introduce horizon-specific measures of overall and marginal variable importance that separate intrinsic model sensitivity from realized sample dynamics, allowing ex ante assessment of which variables truly drive conditional forecasts.
4. **Practical implementation.** All methods are implemented in the open-source R package `cforecast`, designed for applied scenario analysis with reduced-form VARs.

Related: Banbura and Runstler (2011) decompose forecasts using observation weights in factor models. We extend this idea to conditional VAR forecasting with multi-period paths and link it to GIRFs.

Scenario: Risk-off shock with energy disruption

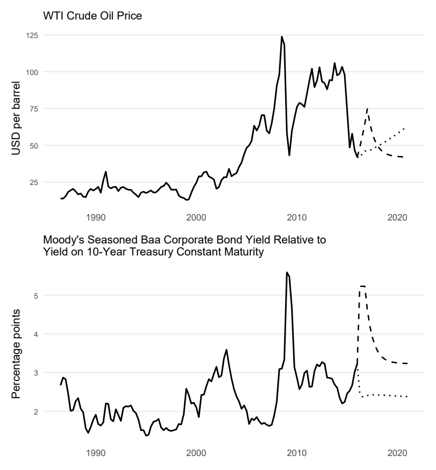
Data and model

- U.S. quarterly data (1986–2015)
- Five-variable reduced-form VAR(2)

Variables: GDP growth, core inflation, policy rate, credit spread (Baa–10Y), oil price

Scenario

- **Credit spread:** +200 bps, persistent, gradual normalization
- **Oil price:** sharp spike, peak in year 1, mean reversion



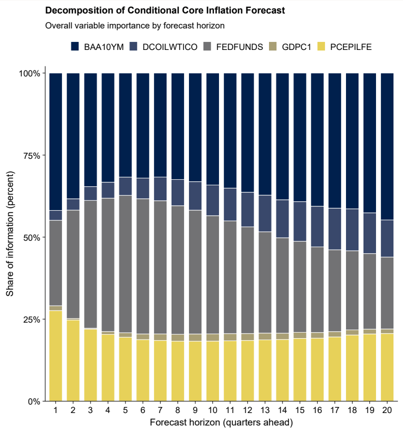
Imposed scenario paths vs. unconditional baseline forecasts.

Ex-ante Variable Importance

Key results

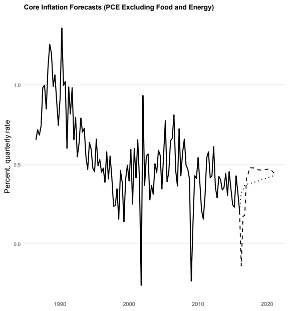
- **Credit spread:** dominant, rising with horizon (Lopez-Salido et al., 2017; Caldara and Herbst, 2019)
- **Oil price:** modest, delayed effect
- **Policy rate:** peaks at 4–6 quarters (Christiano et al., 1996)
- **GDP:** negligible (Stock and Watson, 2007)

Importance computed ex ante (no realized data)

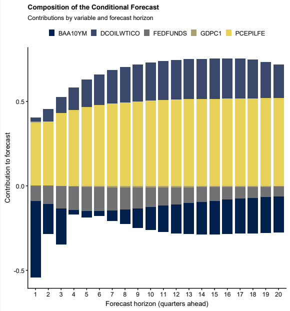


Overall variable importance across horizons.

What Drives the Scenario Forecast?



Conditional vs. baseline forecast



Forecast decomposition

Key takeaways

- Credit spreads are the dominant driver of the forecast revision.
- Oil matters mainly through the imposed path magnitude, not model sensitivity.
- Decomposition identifies which scenario assumptions are pivotal.

Additional Tools for Scenario Interpretation

Further empirical exercises in the paper

Constraint-by-constraint attribution	Identifies which imposed path elements drive forecast revisions.
Constraint-horizon rankings	Ranks which quarters of a scenario are pivotal.
Absolute impact profiles	Summarizes how influence is distributed across horizons.
Pruned-scenario validation	Shows a small subset of constraints can reproduce much of the full scenario.
Capture curves	Quantifies redundancy and supports parsimonious scenario design.

Practical implications

- Interprets what drives the scenario, not merely what the scenario implies.
- Supports simpler, more transparent scenario design and communication.

Methodology

Our approach builds on Kalman filtering and the observation-weight extraction method of Koopman and Harvey (2003).

Consider a VAR(p) for the $n \times 1$ vector y_t:

$$y_t = \sum_{i=1}^{p} \Phi_i \, y_{t-i} + u_t, \qquad u_t \sim (0, \Sigma), \quad t = 1, \ldots, T.$$

A conditional forecast s periods ahead, given an imposed future path and information Ω_T, can be written as[1]:

$$\mathbb{E}[y_{T+s} \mid \text{constrained future path}, \, \Omega_T] = \sum_{i=1}^{T+h} W_{T+s,i}(T+h) \, y_i.$$

[1] Missing observations have zero weights.

GIRF recap (unscaled)

Assuming that the process $\{y_t\}$ is (weakly) stationary, it admits the following infinite-order moving average (MA) representation:

$$y_t = \sum_{i=0}^{\infty} A_i \, u_{t-i}, \quad t = 1, \ldots, T,$$

where the $n \times n$ matrices $\{A_i\}$ are determined recursively as:

$$A_i = \Phi_1 A_{i-1} + \Phi_2 A_{i-2} + \cdots + \Phi_p A_{i-p}, \quad i \geq 1,$$

with $A_0 = I_n$ and $A_i = \mathbf{0}$ for $i < 0$. For a shock to innovation component u_{jt}:

$$GIRF(s, \delta_j, \Omega_{t-1}) = \mathbb{E}[y_{t+s} \mid u_{jt} = \delta_j, \Omega_{t-1}] - \mathbb{E}[y_{t+s} \mid \Omega_{t-1}].$$

Under Gaussianity:

$$\mathbb{E}[u_t \mid u_{jt} = \delta_j, \Omega_{t-1}] = \Sigma e_j \Sigma_{jj}^{-1} \, \delta_j,$$

so

$$GIRF(s, \delta_j, \Omega_{t-1}) = A_s \Sigma e_j \Sigma_{jj}^{-1} \, \delta_j.$$

Path conditioning in reduced-form VAR scenarios

Forecast horizon: $\mathcal{T} = \{T+1, \ldots, T+h\}$.
Stacked restriction on observables:

$$\mathcal{S}\,\mathbf{y}_{\mathcal{T}} = \boldsymbol{\delta}_y^c, \quad \mathbf{y}_{\mathcal{T}} = (y'_{T+1}, \ldots, y'_{T+h})' \in \mathbb{R}^{nh}.$$

- Conditioning is on **future observables**, not on shocks.
- Implemented via state-space + Kalman filtering/smoothing.

In the linear-Gaussian state-space representation,

$$\mathbb{E}[y_{T+s} \mid \mathcal{S}\mathbf{y}_{\mathcal{T}} = \boldsymbol{\delta}_y^c, \Omega_T] = \sum_{i=1}^{T+h} W_{T+s,i}(T+h)\, y_i.$$

- $W_{T+s,i}(T+h)$ are **observation weights** (Koopman and Harvey, 2003).
- Missing observations $\Rightarrow$ corresponding weights are zero.

Conditional Response Function (CRF)

Definition. The conditional response function (CRF) measures the influence of an imposed constraint on a specific forecasted outcome.

For variable j, forecast horizon s, and a constraint imposed at time $T + l$, the CRF is defined as:

$$CRF_j(s, l, \mathcal{S}\mathbf{y}_T = \boldsymbol{\delta}_y^c, \Omega_T) \equiv W^{(j)}_{T+s,\, T+l}(T + h),$$

where $W^{(j)}_{T+s,\, T+l}(T + h)$ is the observation weight assigned to the constrained value $y_{j,T+l}$ in the conditional forecast of y_{T+s}.

- Unlike GIRFs, which trace shocks to reduced-form innovations, CRFs capture the effects of **path restrictions on future observables**.
- Path conditioning imposes weaker—but operationally feasible and often more interpretable—restrictions than structural shock identification.
- Through the MA representation, imposed paths implicitly restrict the joint distribution of future shocks and reshape the entire forecast trajectory.
- CRFs provide a direct, quantitative measure of how each constrained observation influences future outcomes.

CRF and GIRF: Main Results

Proposition 1 (Caspi and Ginker, 2025).

(i) For a single-period conditioning on $y_{j,T+1}$, the conditional response function $CRF_i(s, 1, \delta_y^c, \Omega_T)$ coincides with the *unscaled* generalized impulse response function (GIRF) at horizon s.

(ii) For a general multi-period conditioning over $\mathcal{T} = \{T+1, \dots, T+h\}$, the CRF weight assigned to $y_{j,T+1}$ in the smoothed forecast of $y_{i,T+s}$ differs from the corresponding unscaled GIRF at all horizons s.

Implication: CRFs generalize impulse response analysis to path-based scenarios and form the basis for new scenario analysis tools.

Variable importance for conditioning sets

Let $w_{jk}(s)$ be smoothing weights for variable j, time k, horizon s. Scale by σ_j.
Overall importance (VI):

$$VI_j(s) = \frac{\sum_{k=1}^{T+h} |w_{jk}(s)\sigma_j|}{\sum_{j=1}^{n} \sum_{k=1}^{T+h} |w_{jk}(s)\sigma_j|}.$$

Marginal importance (MI) from imposed future paths only:

$$MI_j(s) = \frac{\sum_{k=T+1}^{T+h} |w_{jk}(s)\sigma_j|}{\sum_{j=1}^{n} \sum_{k=1}^{T+h} |w_{jk}(s)\sigma_j|}.$$

Thank you!

Questions and comments are welcome.